

Do Carebots Care? The Ethics of Social Robots in Care Settings

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00:00

Hi, I'm Shannon Vallor. I'm the Baillie Gifford Chair in the Ethics of Data and Artificial Intelligence at the University of Edinburgh. And this is a presentation on AI ethics, and specifically on the topic of care bots. And we're going to ask and try to answer the question, do care bots care? And what are the ethics of using social robots like care bots in care settings. So let's get started. I'm going to share my slides.

00:37

So, let me give you some background about me. I direct the Centre for Technomoral Futures at the Edinburgh Futures Institute at the University of Edinburgh, where I'm also appointed in the Department of Philosophy. I came to Scotland in 2020 from California where I had worked in Silicon Valley for over a decade as a philosopher and an ethicist of emerging technologies. I've had experience collaborating with industry, with government, with policymakers, and with technologists across the spectrum of computer science, engineering, robotics, data science, to try to develop tools, techniques, knowledge, concepts and practices that help technologists embed ethical practices, ethical understanding into the design, development and deployment of new technologies.

01:45

Today, we're going to talk about something called robot ethics. Now, in this context, we're not talking about the ethics that robots might have. We're talking about the ethics that people who design, develop and deploy robots have to apply in order to do that responsibly. So we're talking about the ethical implications of robots, and today we're going to focus on social robots, and specifically, a subset of those called care bots, social robots that are used in care settings. This is a new area of interdisciplinary study, and it draws from many different disciplines - engineering ethics, AI ethics, data ethics, philosophy, computer science, mechanical engineering, psychology, anthropology, sociology, human-robot interaction, and human-machine interaction as interdisciplinary fields, and art and design.

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The core issues in robot ethics are many, we're gonna focus on several of these today. But we will never be able to capture or exhaust the full list of ethical issues that social robots can present, in part because we can't anticipate the new issues that we will face as the technology develops, and as new applications emerge. But in general, we focus a lot on human safety and non-harm, issues of privacy, surveillance, and control of robots, the kind of security that is necessary in order to make sure that the

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data that robots collect is not exfiltrated or abused. The way that robots themselves are designed, whether they're designed with the inclusion of affected populations, whether they're designed by sufficiently diverse teams, and whether the applications of robots represent humans in ways that are fair, and just. We look at the way that robots affect the workplace and the future of labor; how humans and robots interact, and what forms of relationships and dependencies can emerge from humans to robots. We look at the ways that implementing robots, especially social robots, can affect human dignity and autonomy, human freedom to act and decide in our worlds as we choose. We look at affect of robotics, the ways that robots are increasingly being designed to both detect human emotions and to simulate emotions in return, and the ethical implications that these present are immense. We have questions about how sustainable robotics technology is, how responsible the sourcing of the materials used in robots is, how disposable and environmentally friendly or unfriendly the materials used in robotics are; and whether robots are used in ways that are just, accountable, transparent and explainable to humans.

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There are a lot of different commitments within the broader field of technology ethics that robot ethics draws upon. First of all, we have to recognize that technologies, including robots are never value neutral. They're always designed with human purposes, and intentions, and goals, and values in mind. We have to recognize that technologies also do many things - that's what we mean when we say they're polypotent. So we don't only need to think about the design's intent - what we hope the robot will do or what we intend the robot to do - we also have to think about the other things that might be used to do that we didn't intend or anticipate. We have to recognize that technologies like robots are always co-shaped by users, so the designers don't have complete control over what the technology will be used to do. We also have to recognize the technologies always embed and reflect the societies and cultures in which they originate, and the values are shaped by those cultures. We have to think about the fact that responsible robot development requires ethical work, and that this goes beyond good intentions or complying with the law and policy that's applicable in the domain in which we're working. And that means we have to grapple with moral uncertainty, value trade offs, value conflicts, and what we call wicked problems, problems where there is no optimal solution that presents no risk, no harm, and that satisfies all of the moral or ethical values that we would like to promote. So tech ethics, and by extension, robot ethics is an ongoing responsibility of technologists. It's not a box to be checked off. So the short version of this is there's no algorithm for ethics, no algorithm for technology ethics, and definitely no algorithm for robot ethics. This is an ongoing practice, not a problem that will be solved. So how does this apply to the use of social robots in care settings, what now goes by the name care bots? Well, let's look at some examples of care bots. So one possible application, one that's actually already well-developed and in some commercial use and additional testing, is the use of robots in homes, care homes and hospitals for elder care. We might also extend this to care for those who are vulnerable in other ways, people who are, for example, recovering from strokes, people who are facing chronic diseases that impact their ability to live independently. So when we talk about these kinds of care bots, we might be thinking of companion robots. One popular commercial robot in nursing homes is a mechanical seal. You can see him in the background of this image, a mechanical white seal called Paro, and Paro has been quite successful in nursing homes as a early iteration of a care bot. Paro is a mechanical seal that makes noises when you touch it, that moves around, that seems to respond positively to affection, and petting and attention. And we've seen that Paro has had some positive

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effects on residents of care settings. We also have robots that primarily perform tasks of health monitoring, robots that can take your temperature or your blood pressure, or assess your mobility, making sure that you, for example, get up in the morning and take your pills. We have robots that function as telerobots for remote care, meaning it's a robot that is mobile, that can be piloted by a human user who can appear on the screen. So a physician in another location might use a telerobot to come into a patient's room and check on them, or a resident in a nursing facility might be checked on by a family member by means of a telerobot. And we're also looking at the development of care bots to perform physical tasks like lifting patients, turning them in bed, tending to intimate needs, or helping with physiotherapy.

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There are also potential care bot applications with children in homes, daycare and schools. And obviously, this is a tremendously ethically sensitive and risk bound approach. But nevertheless, there are already plenty of pilot studies and examples of robots being used in schools as companion or play robots. We have many examples of robots in development as therapy robots, for example, for children with autism spectrum disorder, or other conditions where there is a value for robotic therapy. Telerobots, as described in the previous slide, can also be used in schools for remote learning. They can also be used to assist teachers in the classroom, or they can perform tasks like classroom monitoring. One question we have to ask is whether care bots care? What do we mean when we ask that question? Well, there are important limits to robot care that we have to acknowledge. Care bots can perform some caring tasks. But keep in mind that robots, even when they imitate human emotion, actually lack the emotions and motivations that stand behind human care. So a robot that appears to be moved by an expression of pain or discomfort in an elderly patient, for example, and moves forward to assist or to get help, is not actually caring in the emotional sense, even if they display signals of emotion. So robots don't actually understand the meaning of care or its moral and social significance. They are merely machines that are programmed to respond in ways that support caring tasks, but they don't have the cognitive or emotional dimensions of care that human caregivers possess. This means that they can't recognize new or unexpected care needs that they were not designed to detect. And it also means that they're not yet able to safely perform many tasks of physical caregiving. And this is true both because of the limitations or absence of understanding of the significance and risks of care, but also because from a biomechanical standpoint, it is very difficult to design robots to perform physical tasks that carry no meaningful risk of injuring human patients. There are also some risks of robot care that we have to take into account. Despite the fact that robots have no genuine emotions, humans can form strong emotional attachments to social robots. And these attachments, as we've said, will not be reciprocated. And yet, the robot design might mimic reciprocation of affection, or love, or friendship, furthering these attachments, and we have to ask about the potential benefits and harms of these attachments. Humans can also be deceived or manipulated by simulations in robots of emotion and caring intent. And there are risks that are associated with this as well as questions about the inherent ethics of deceiving or manipulating people, especially if they are in a vulnerable population, such as children or the elderly. Humans in some cases may actually form a preference for robot care over human care or companionship. And we have concerns about whether that might diminish human interaction in cared for populations. And finally, humans can overestimate the capabilities of robots that appear intelligent or sentient, meaning that they may rely upon the robot to respond intelligently in a

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risky situation where the robot may not be designed to actually understand and respond appropriately to that particular context - that can present safety risks.

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What are the advantages of robot care? Well, they can increase access to caring services for many vulnerable and isolated persons. They can stimulate and sustain human cognitive, emotional and communicative capabilities. And they can even promote more human-to-human interaction than in care settings without robots. And there's considerable empirical evidence of pilot studies of robots in care settings that suggest that robots can indeed have these benefits. They can in fact, stimulate humans to interact more with the humans around them through the mediation of the robot, and they can activate human cognitive, emotional and communicative capabilities that might otherwise decline in care settings. Social robots can also provide a valuable sense of privacy and control for many people in care settings. For example, in some contexts, patients have indicated that they prefer the presence of a robot assistant in settings where there might be a potential embarrassment or fear of exposure to human judgment, because residents or patients might be aware that the robot is incapable of moral judgment, or any kind of intrusive observation of intimate or embarrassing details that might emerge in a care setting.

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So when is using a care bot ethical? Well, I'm going to present a thought experiment for us to consider. So this is an entirely hypothetical robot. This robot doesn't exist, but we're going to imagine it. So consider a problem that is well-known in the care of moderate to advanced dementia patients. Such patients often suffer from emotional dysregulation - that is unprovoked anger, frustration, suspicion, confusion, fear or anxiety that can both present obstacles to caregivers, and also decrease the quality of life for the dementia patient. So let's envision a robot designed to help with this challenge. We'll call this robot Cessabot, which is a play on the Latin word for calm. So imagine that this is a social robot designed to help residents of memory care facilities with their emotional regulation, that is to help residents with the management of their own emotional states. Keep in mind that human caregivers such as nurses care, home assistants and family members are often overburdened and unable to help patients with their emotional regulation. And when they do, it often increases caregivers own emotional exhaustion and risk of burnout. So Cessabot might be helpful not only for dementia patients, but for the humans who provide those patients with primary care. So imagine, then that you have a hypothetical AI company that advertises this robot, it will work directly with residents using AI and machine learning for sentiment analysis. So that's the technology that will allow it to understand or rather predict and classify the emotional states of the residents. And it will also have affective robotics design, meaning the robot itself will be able to simulate emotional states. So with the combination of these technologies, sentiment analysis on the one side, and the ability to imitate or mimic effect or emotion itself on the other, the robot will be able to detect negative emotions in residents such as fear, anxiety, anger, grief, confusion. It will be able to engage residents in calming behaviors and interactions. The robot will itself mimic displays of positive and balanced emotions in order to allow residents to mirror those states themselves. And the idea is that over time, Cessabot should be able to steer residents toward calmer and more stable emotional lives.

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So here's some ethical considerations for us. Is Cessabot deceptive, or manipulative? Cessabot is pretending to express emotional states that Cessabot lacks, Cessabot can only mimic or imitate emotion. So is that deceptive? Is it manipulative? Considering that the intention of the simulation is to elicit similar displays from the resident to change their emotional state? Is it unethical to deceive or manipulate dementia patients in order to assist them with emotional regulation? What should the residents or patients be told about Cessabot? Should they be given all of the information about what Cessabot is for? Should they be told what Cessabot is incapable of? Should they know, for example, that Cessabot is not capable of genuine emotional states? What value assumptions are implicit in Cessabot's design? For example, what are the values of positive or negative emotional displays that the designers of Cessabot have? And are they well aligned with patient benefit? Can a robot that doesn't understand or experience emotion ever be trusted to regulate our emotions? Is it appropriate to put a machine that has no emotional life in a role of regulating our emotional lives? And how could known problems for example of racial and gender bias in AI and machine learning training data and model performance - for example, problems of bias in facial recognition, sentiment classification and so forth, result in Cessabot giving unfair, ineffective or otherwise harmful treatment to certain groups of residents. So for example, what if Cessabot does not perform as well in regulating and properly classifying the emotional states of women as it does to male patients? Or what if it is unfair and ineffective in its treatment of patients from minority ethnic groups or different cultural backgrounds that might have very different emotional signaling protocols or norms in their communities? How might designers anticipate and prevent some of these harms? What safeguards or human oversight would need to be in place? For something like Cessabot to be a responsible and beneficial application? What input should care workers have in Cessabot's design and use? What harm can Cessabot do if it performs other than as expected? Is it open to abuse or what we call dual use, where it could be applied for a different purpose, one that is explicitly harmful? Is a care bot such as Cessabot the best available solution or tool for addressing this problem? Have non-technological solutions or remedies been given adequate consideration? These are all the kinds of questions that we should ask in any similar application or scenario where a care bot is being deployed.

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So let's give an overview of some of the high-level issues that we address in care bot ethics. We've talked about safety and non-harm. And here we have to consider both the physical safety of humans and their psychological safety. We also always have to focus when we develop social robots on the potential harms to the most vulnerable persons and groups. And again, we may be dealing with inherently vulnerable populations, such as children or the elderly. But within those groups, there are likely to be sub-populations that are marginalized or minoritized, or otherwise subjected already to greater social vulnerability. And those are the individuals and groups that we have to take most care to protect. We also have to maintain human oversight and meaningful human control of social robot behavior, and ensure that robot functions and goals remain aligned with human interests and values. There are also considerable risks around privacy, surveillance, consent and control that have to be considered. It's important to think about third-party and non-user privacy as well. Of course, we have to consider the privacy rights of residents or those who are being cared for by care bots. But think about, for example, visitors to a care facility, who are not part of the care team nor patients. And yet the robot might be observing and interacting with them. How do we ensure that their privacy and rights are protected? Also think about who has an interest in obtaining behavioral data collected by social robots

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beyond the intended groups such as caregivers, and medical professionals. Think about the interest that law enforcement or employers or insurers could have in this kind of data. Think about the fact that informed consent is often a challenge or sometimes impossible with vulnerable populations. So when is proxy consent, consent from another responsible party, not only legal but ethical? Can users of social robots control their exposure? Can they withdraw or refuse to engage with a robot without facing any harmful consequence or loss?

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We also have questions of justice, fairness and accountability to consider. Who benefits from using this robot directly or indirectly? Who will be put at risk? Are those who stand to gain the most from the use of a robot also the ones most exposed to the risks? Has the latter group, the group who might be at risk from this technology been involved in its design? They should be. How might bias be baked into the design, or the model or its training data, or the use case itself? What kinds of existing social unfairness could a given robot amplify or perpetuate? And who is accountable for assuring the robot's proper use and reliable operation? Who is accountable if someone is harmed by the robot? We mentioned earlier, it's important to think about the sustainability of robots. What's their carbon footprint? Are their parts sourced from ethical supply chains? What is the lifecycle of the robot? Is it easy to fix? And will they lead to a substantial increase in e-waste or plastic waste? There are many considerations about security and abuse and dual use of these technologies. What data from this robot might be vulnerable to attackers or other third-party interests? What protocols are in place to prevent this? What malicious or unintended purposes could the robot or its data be made to serve if it were hacked or otherwise misappropriated? And what adversarial or unintended inputs could make this robot behave in undesirable ways? We have questions about labor in the workforce. Does this robot aid or support human labor or does it replace it? There is an important ethical difference there. If it replaces human labor, what justifies this? Is the work itself inherently dull, dirty or dangerous for humans? Does this robot labor make the service or end product better? In the case of care bots, does it make care better or just more profitable? Will the robot foster economic health or destabilize a local or regional economy? And how will this robot impact workplace rights or quality of work for humans? So for example, in the care bot context, how will the care bot improve the quality of employment and the safety and health of the workplace for human caregivers and workers? We also have questions about transparency, explainability, and human agency. Are all the stakeholders clear about what the robot does and why? Can the purposes or operations of the robot be seen as hidden, manipulative or deceptive? Does it interfere with or diminish human agency or autonomy, the ability to make our own choices and govern our own lives? Does the robot nudge us into behaviors that we would not freely choose? Is the robot's use consistent with respect for human rights and dignity? And again, this is all the more important when we're talking about robots designed to interact with vulnerable human populations. We also have to think about accessibility. Was the robot designed with the abilities and bodies of all users in mind? Will the robot make certain activities more challenging or costly? And again, we have to consider the non-users who will encounter the robot. If designed to aid accessibility or unique needs, has the target user group been actively involved in design, development and testing? For example, in the care bot context, think about a mobile social robot that is moving around a care home. Has it been designed to anticipate the interaction with users of wheelchairs, walkers, or individuals who might have non-standard mobility in the setting? This is also part of a broader ethical strategy of representation, inclusion and participation. How does the robot's design and presentation

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engage issues of gender, race, age, nationality and other identities? We have to remember that robots are never neutral. They're always designed from a perspective that takes certain goals to be priorities, that takes certain values to be important to foreground and promote. And even design choices like the color of a robot are not, in fact, neutral. They represent norms, expectations that often need to be challenged. So what stereotypes or constraints might a given robot reinforce in society, or in a particular setting? Who had a real opportunity to shape the robot's design and development? Who was not merely in the room, but actually given a voice and a platform to shape that design? And one way of summarizing this responsibility is to ask the question, does this robot conform to or violate what has become a new ethical norm of technology design - "nothing about us without us"? This is a phrase that arose in the realm of disability studies and activism. The idea that technologies for disabled people should never be designed, developed or deployed, without the full participation and involvement of disabled people themselves. But this can be broadened to other populations, especially vulnerable populations as well. And think about the ways that robots affect our relationships with one another, and with the world around us. Could people be hurt by dependence on this robot's labor or its social presence? Will humans know when not to rely on the robot, and what its limits are? Will the robot project capabilities that it doesn't have therefore deceiving us, for example, a robot that says "I love you"? And will robots be tempted to sabotage or destroy the robot? How will the robot impact caring bonds with other humans and animals in a given setting? These are essential questions to ask.

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So remember, what is the too long don't read summary? There's no algorithm for robot ethics. There's no algorithm for care bot ethics. It's not a problem that we solve. Ethics is a tool that we pick up and use every day. And it's a vital part of responsible robotics, design, development and use. Thank you. And here's some further reading for you that you may find useful if you'd like to dive deeper into this topic. Thanks.